MASTER'S THESIS PROPOSAL

Spatiotemporal Modeling of Citizen Tree Watering in Berlin

Weather Responses and Neighborhood Variation

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Abstract

Berlin's heat burden is spatially uneven, and the tree canopy that moderates it does not deliver a constant service: evapotranspirative cooling weakens when soil water is limited, which is when heat is most dangerous. Since May 2020, the citizen platform "Gieß den Kiez" has logged about 59,000 volunteer tree-watering events with date, location, tree identity and volume, which provides a geolocated, dated record of a voluntary climate-related behaviour across a whole city over six summers, including two drought years.

This thesis has one core study and two supporting studies. The **core study (Part II)** models daily logged watering in Berlin's 542 planning areas as a count response to heat and dryness, yielding a neighbourhood civic heat-response indicator. It tests whether that response is weaker in socially disadvantaged areas, and whether any such gap is explained by participation inequality rather than vulnerability. **Part I** involves the estimation of how much surface cooling from canopy and low vegetation is lost on dry days and where that loss falls, extending recent US evidence to a European city. **Part III** is an exploratory decision-support analysis that compares need-only and complement-based allocations of municipal irrigation for young street trees. A bridging test asks whether the civic response follows cooling loss.

Outcomes are stated precisely: Landsat morning land surface temperature (LST) in Part I, logged watering counts and litres in Part II, and expected unmet water and allocation rankings in Part III. None of these is treated as felt heat or as the causal effect of watering. All data are public, apart from two optional requests to CityLAB and the district green-space offices.

1. Introduction and Motivation

Heat exposure in Berlin is amplified by dense built form and night-time heat retention, and older residents and other vulnerable groups face disproportionate risk (Burkart et al., 2014; Gabriel and Endlicher, 2011; Masselot et al., 2023). Street and park trees are therefore both ecological infrastructure and a public-health adaptation resource.

Two facts about that resource are usually stated separately. First, tree cooling is inconstant. Since it is driven by evapotranspiration, it weakens when soil water is limited (Schwaab et al., 2021; Gobatti et al., 2025). Second, the maintenance that carries trees through dry periods is growingly shared. Berlin's district green-space offices water newly planted street trees for roughly their first five years. After that, much of the watering is rather on a voluntary basis.

The Gieß den Kiez (CityLAB Berlin, CC Zero) program records these voluntary waterings, and the Baumkataster (Tree register) lists more than 950,000 catalogued trees in Berlin. Together, they allow

recorded civic behavior to be analyzed alongside environmental measurement, rather than as a descriptive engagement metric.

The "obvious" study on whether watering *measurably* lowers local LST is not attempted. Appendix A shows that at observed watering densities, the expected surface signal is well below what Landsat-class sampling can resolve, and that the poured volume mostly infiltrates rather than evaporating at overpass time. The thesis instead asks who responds to heat, where cooling is lost, and how scarce municipal effort could be allocated around volunteers.

The thesis prioritizes multi-source geospatial data engineering, hierarchical inference with partial pooling, leakage-resistant spatial and temporal validation, and a transparent decision layer.

2. Background and Related Work

2.1 Conceptual framework

The thesis uses the IPCC risk framework (IPCC, 2014): heat risk arises from hazard (high temperature), exposure (people and assets in hot places) and vulnerability, which combines sensitivity with adaptive capacity (Smit and Wandel, 2006). Each variable is displayed in Table 1. Adaptive capacity is a latent potential, as the watering record observes realized action. The thesis therefore treats civic heat response as a behavioral indicator relevant to adaptive capacity, and tests how far the two coincide.

Framework element	Operationalization in this thesis	Source
Hazard	Daily maximum air temperature; antecedent dry-spell length; drought anomaly	DWD, ERA5, ESA CCI
Exposure (surface)	Morning land surface temperature; baseline thermal load	Landsat 8/9 C2 L2
Sensitivity	Share of residents aged 65+; social status index	Einwohnerregister; MSS 2025
Adaptive capacity (stock)	MSS social status index; Environmental Justice Atlas (secondary)	Berlin Senate
Behavioural response	Civic heat-response slope of logged watering	Gieß den Kiez
Ecosystem service	Per-unit and total vegetation cooling; its loss on dry days	Landsat, Sentinel-2, Umweltatlas

Table 1. Framework elements and their operationalization.

2.2 Vegetation cooling and drought

Trees lower LST substantially across European cities, with effects that vary by regional climate (Schwaab et al., 2021; review in Alonzo et al., 2025). Soil moisture limits evaporative cooling (Gobatti et al., 2025). Even during the 2022 drought, airborne thermal imaging in Bavaria showed tree canopies 2–6 °C cooler than air, with strong species differences and modulation by imperviousness (Zandler and Samimi, 2024). This implies that canopy and low vegetation respond to drought very differently, a distinction that NDVI cannot make alone.

2.3 Distributional incidence of drought-degraded cooling

The incidence of this loss has recently been studied in the United States. Across 45,254 census tracts in 245 cities, Yan et al. (2025) found that vegetation declined more during droughts in communities with higher shares of people of color, and that daytime cooling fell from 0.59 to 0.49 °C per 0.1 NDVI; they attribute much of the disparity to unequal access to irrigation water. Whether this pattern holds in a European city where public crews water young street trees, and at planning-area resolution, has not been tested.

2.4 Measuring adaptive capacity and behavioral heat response

Adaptive capacity is conventionally measured with static composite indices built from census stock variables (Cutter et al., 2003). Behavioral measures of heat adaptation already exist: subway ridership in New York City responds less to heat in low-income neighborhoods (Stechemesser and Wenz, 2023), and mobile-phone mobility data show unequal responses to heat in US cities. These studies observe individual protective behavior. No study we have found observes a collective, civic response, i.e., residents acting on shared urban infrastructure, at a daily resolution across a whole city.

2.5 Civic stewardship and participation inequality

Co-production of public services depends on citizens' time, skills, and access (Ostrom, 1996). Reporting of public-space problems through digital channels is unevenly distributed across neighborhoods (O'Brien, 2018), and platform design itself shapes who participates (Kirklies et al., 2024). Street-tree stewardship also varies with socio-economic context. For this thesis, the key implication is a rival explanation: weaker watering response in disadvantaged areas could reflect unequal platform awareness, digital access, or logging habits rather than lower adaptive capacity. Part II is designed to test this rival explanation.

2.6 Decision support for tree irrigation

Data-driven irrigation planning is advancing. Berlin's QTrees project combines soil-moisture sensors and machine learning to estimate street-tree water needs (CityLAB Berlin), and Frankfurt has built an IoT-based irrigation decision-support system for 8,000 young trees (Rebholz et al., 2026). These systems estimate how "thirsty" trees are, not accounting for where volunteers have already watered. Part III thus focuses on allocation around volunteer effort and is designed so that sensor-based water-need estimates can replace its simple need term when they become available.

3. Research Gap

Each gap below is stated only as far as the literature in Section 2 supports it.

- **Gap 1 — Incidence of cooling loss in a European city.** Unequal drought-driven cooling loss is documented across US census tracts (Yan et al., 2025) but has not been estimated within a European city, where street-tree irrigation is largely public, nor separately for canopy and low vegetation.
- **Gap 2 — Collective civic response to heat.** Behavioral heat-adaptation studies observe individual mobility. A daily, city-wide record of residents acting on shared green infrastructure has not been used to measure how neighborhoods respond to heat, or how that response relates to static indicators of capacity.
- **Gap 3 — Allocation around volunteer effort.** Existing irrigation decision support estimates tree water need; we found no published method that allocates municipal irrigation around observed volunteer effort. This will be confirmed by a structured literature search in Month 1.

A further, secondary gap is the persistence of volunteer care: 78% of watered trees were watered in only one of six seasons and mean daily summer events fell from 121.6 (2020) to 19.4 (2024). The thesis reports persistence descriptively as context for all three parts.

4. Research Questions and Hypotheses

4.1 Structure and priorities

Study	Role	Question in brief	Share of effort
Part II — Civic heat response	Core	Who responds when it is hot, and why does response differ?	≈ 45%
Part I — Cooling loss	Supporting	Where is vegetation cooling lost on dry days?	≈ 30%

Study	Role	Question in brief	Share of effort
Part III — Allocation	Exploratory	How do need-only and complement-based allocations differ?	≈ 15%
Bridge (H4)	Integration	Does civic response follow cooling loss?	≈ 10% (with synthesis)

4.2 Research questions

- **RQ1 (Part I).** In Berlin, 2017–2024, how much weaker is the per-unit relationship between vegetation and morning LST on dry than on wet clear-sky days, separately for tree canopy and low vegetation, and is the resulting total cooling loss per planning area larger where social status is lower and baseline thermal load is higher?
- **RQ2 (Part II).** Does logged citizen watering respond to daily heat and dryness, and does the strength of that response differ systematically with neighbourhood social status after accounting for baseline participation?
- **RQ3 (Part III).** How accurately can logged watering of young street trees be predicted under spatial and temporal hold-out, and how much do need-only and complement-based allocations differ in which planning areas they prioritize?

4.3 Hypotheses

All four hypotheses, their test statistics and decision thresholds will be pre-registered (OSF) before outcome data are modelled.

- **H1 — Unequal cooling loss.** Total cooling loss on dry days (loss in per-unit cooling × local vegetation fraction) is larger in planning areas with lower MSS social status and higher baseline LST. *Test:* sign and 95% credible interval of the greenness × drought × status interaction; reported separately for canopy and low vegetation.
- **H2 — Unequal response.** The heat- and dryness-response slopes of logged watering are lower in planning areas with lower social status. *Test:* coefficient of status on the random slopes within the hierarchical model.
- **H2-alt — Participation inequality.** The status gradient in H2 is explained by differences in baseline participation and digital-access proxies. *Test:* H2 is supported over H2-alt only if the status effect on slopes remains after conditioning on the planning-area intercept (baseline participation), the older-population share and the turnout proxy described in Section 6.4.
- **H3 — Divergence from the static index.** Civic heat response and the MSS social status index agree only weakly. *Test:* Spearman rank correlation between posterior slopes and the status index below 0.5, and at least 20% of planning areas in off-diagonal tercile cells. The Environmental Justice Atlas classification is reported as a secondary comparison, because it measures environmental burden rather than capacity.
- **H4 — Care does not follow cooling loss (bridge).** Planning-area civic heat-response slopes are uncorrelated or negatively correlated with total cooling loss from Part I. *Test:* Spearman correlation with posterior uncertainty propagated from both parts.

4.4 Analytical mapping

Question	Primary method and decision rule
RQ1 / H1	One-stage hierarchical model of Landsat LST on 30 m product-grid cells: vegetation (canopy, low vegetation) × temporal drought anomaly × planning-area status, with date fixed effects and planning-area random slopes. Placebo on non-vegetated cells. Reported in °C per 10 percentage points of canopy or low vegetation.
RQ2 / H2 / H2-alt / H3	Two-part model. (a) Whether a planning area participates at all (logistic, area level). (b) Among participating areas, hierarchical negative-binomial model of logged events per area-day with an

Question	Primary method and decision rule
	offset for young street trees, random intercepts and random slopes on standardised Tmax and dry-spell length. Status, older-population share and turnout enter as predictors of intercepts and slopes.
RQ3	Regularised logistic regression and gradient boosting on tree-seasons of young street trees; spatially blocked (planning area) and temporal (train 2020–2023, test 2024–2025) validation; AUROC, PR-AUC, isotonic calibration, Brier score. Allocation compared by Spearman rank correlation and top-50 overlap.
H4	Spearman correlation between posterior draws of Part II slopes and Part I total cooling loss.

5. Data Sources

"Observation unit" is the delivered spatial or temporal unit of a value, which is not always the instrument's native resolution. Landsat LST is delivered on a 30 m grid from about 100 m native thermal sampling.

Dataset	Provider	Use	Unit / limit	Part
Landsat 8/9 C2 L2 LST, 2013–2024	USGS via Earth Engine	Outcome (Part I); thermal load	30 m grid; ~100 m native; clear-sky only; ~10:30 overpass	I, III
Sentinel-2 L2A NDVI, 2017–2024	ESA Copernicus	Greenness, dated	10 m, aggregated to 30 m	I
Umweltatlas Vegetationshöhen 2020	Berlin Senate	Canopy (>5 m) vs. low vegetation fraction	1 m raster, single date	I, III
ECOSTRESS LST (where available)	NASA	Afternoon LST sensitivity check	~70 m; irregular times	I
Imperviousness 2021 (2016 sensitivity)	Umweltatlas	Sealing covariate	Polygon / raster	I, III
Land use, urban structure, waterbodies	Berlin Geoportal	Built-form controls; distance to water	ISU5 polygons	I, III
Baumkataster	Berlin Geoportal	Tree population, species, planting year; offsets	One record per tree; green-space coverage incomplete	II, III
Gieß den Kiez logs	CityLAB, CC Zero	Outcome (Part II); label (Part III)	Event: date, litres, tree ID; 2020 from late May	II, III
DWD station data and heat warnings	DWD CDC	Tmax, precipitation, dry spells, heat-warning days	Station-day	I, II
ERA5; ESA CCI Soil Moisture v09.2	Copernicus; ESA	Temporal drought anomaly and alternatives	0.25°, city-level only	I
Monitoring Soziale Stadtentwicklung 2025	Berlin Senate	Primary status index	Planning area; some exclusions	I–III
Environmental Justice Atlas 2023/24	Berlin Senate	Secondary benchmark (H3)	Planning area	II
Einwohnerregister	Amt für Statistik BB	Population, 65+ share	Planning area, annual	I–III
Election turnout (2021, 2023)	Landeswahlleiterin Berlin	Civic-participation proxy (H2-alt)	Electoral district → planning area	II
School holidays; public holidays	Berlin Senate	Calendar controls	Day	II
Requested: pseudonymous user IDs	CityLAB	Breadth vs. depth of participation	Event-level, if granted	II
Requested: district watering schedules	Bezirksämter	Municipal care status of trees	Tree or street, if granted	III

6. Methodology

6.1 Study area and units of analysis

The study covers Berlin's 12 districts (about 892 km²). Units match the support of each dataset: 30 m Landsat cells aggregated to planning areas (Part I), 542 LOR Planungsräume with 143 Bezirksregionen as fallback (Part II), and individual trees aggregated to planning areas for allocation (Part III). Adjacent 30 m thermal cells are not treated as independent. All area-level results are repeated at Planungsraum, Bezirksregion and district scale (Openshaw, 1984); conclusions are labelled robust only where direction and ranking survive.

6.2 Data audit and signal construction (Month 1)

The logs are deduplicated (111 exact duplicates), screened for implausible volumes (5 events above 500 L), and corrected for swapped coordinates (56 records); 125 records remaining outside Berlin are dropped. Events join the Baumkataster by tree ID at a 98.95% match rate, and 955,434 of 956,347 trees (99.90%) are assigned to planning areas. The audit will also:

- reconcile the three counts of watered trees in the current exploratory work (11,087; 11,160; 11,245) and document the filter behind the final figure;
- define the **at-risk tree stock**: street trees aged 5–19 years, past municipal establishment care but still young, as the primary offset and prediction population;
- flag planning areas with few catalogued trees or weak registry coverage, which are excluded from inference and reported separately;
- record the outcome of the CityLAB and district data requests, which decides whether the concentration and municipal-status analyses run in full or as proxies.

Two properties of the series are fixed in advance. **Overdispersion:** across 1,039 active summer days, daily events have mean 54 and variance-to-mean ratio 93.5, so a Poisson model is ruled out. **Engagement decline:** mean daily summer events were 121.6 (2020), 44.7 (2021), 93.1 (2022), 39.3 (2023), 19.4 (2024) and 24.7 (2025). Year effects and a day-of-season smooth are therefore mandatory, and the identifying variation is within-area, within-year response to weather.

6.3 Part I — Cooling loss and its incidence

Outcome and sample. Landsat 8/9 LST on valid clear-sky summer overpasses, 2017–2024 (2013–2024 for descriptive maps). Clear-sky sampling favours hot, dry, anticyclonic days; inference is limited to that regime, and the weather distribution of observed versus all summer days is reported.

Vegetation. Canopy fraction and low-vegetation fraction per cell come from the 2020 vegetation-height layer; Sentinel-2 NDVI from the nearest valid scene (window length stored) captures dated greenness. Because heat and water stress lower NDVI, lagged NDVI is used as a reverse-causality check.

Drought. A city-level temporal anomaly from ESA CCI soil moisture, with ERA5 and DWD-based SPEI as alternatives. All spatial variation comes from vegetation and built form, never from the coarse moisture grid. Dry and wet are defined as anomaly terciles, with quartile and continuous versions as sensitivity.

Model. A single hierarchical model replaces the earlier extract-then-regress design, so that uncertainty in cooling loss carries through to the equity test:

$$LST(c,t) = \alpha(t) + \beta_1\, V(c) + \beta_2\, V(c)\cdot D(t) + \beta_3\, V(c)\cdot D(t)\cdot S(j) + \gamma' X(c) + u(j) + v(j)\cdot V(c) + w(j)\cdot V(c)\cdot D(t) + \varepsilon(c,t)$$

where c is a cell, t an overpass date, j the planning area, α(t) date fixed effects absorbing common weather and sensor conditions, V vegetation (canopy and low vegetation as separate terms), D the drought anomaly, S social status, and X imperviousness, built density and distance to water. Per-unit cooling loss is the dry-minus-wet change in the vegetation slope; **total cooling loss** multiplies it by the

area's vegetation fraction, and both are reported. Residual spatial dependence is checked with global and local Moran statistics (Anselin, 1995) and handled with a spatial random effect.

Checks. (1) Placebo: the same interaction on fully sealed cells should be near zero. (2) Afternoon ECOSTRESS scenes, where available, test whether the pattern holds closer to peak heat. (3) Imperviousness, built density and vegetation enter separately before the full model, with variance-inflation and condition-index diagnostics. (4) Spatially blocked versus random folds quantify spatial leakage (Roberts et al., 2017).

6.4 Part II — Civic heat response (core study)

Outcome. Logged watering events per planning-area-day, April–September 2020–2025 (2020 from late May); logged litres as a secondary outcome. The full panel is 542 areas × about 183 days × 6 seasons ≈ 595,000 area-days.

Two-part structure. Eighty planning areas record no watering in six seasons. A response slope is not identifiable for them, and zero use may reflect no eligible trees, no awareness, or no capacity. Part (a) therefore models whether an area ever participates (logistic, area level, on status, 65+ share, at-risk tree stock and turnout). Part (b) models the daily response among participating areas.

Daily model. Hierarchical negative binomial with an offset for the log of at-risk young street trees (all catalogued trees as sensitivity), planning-area random intercepts, and correlated random slopes on standardized daily Tmax and antecedent dry-spell length. Fixed controls are daily precipitation, DWD heat-warning days, weekday, school and public holidays, year effects and a day-of-season smooth. Status, 65+ share, population density and turnout enter as area-level predictors of both intercepts and slopes. Slopes are reported as percentage change in expected events per standard deviation of Tmax (semi-elasticities).

Estimation. Models are first fitted at Bezirksregion level and with glmmTMB or INLA for speed, then in brms/Stan (Bürkner, 2017) at planning-area level for final inference, so that thin areas are shrunk toward the city mean (Gelman and Hill, 2007). Posterior means and 90% credible intervals are reported for every area.

Testing H2 against H2-alt. The model estimates the correlation between intercepts (baseline participation) and slopes. H2 is supported only if the status effect on slopes holds after conditioning on the intercept, 65+ share and turnout. If pseudonymous user IDs are obtained, slopes are re-estimated on the number of distinct users per day and on events excluding the top 5% of users, to separate breadth from a few highly active users. Without user IDs, distinct trees watered per day and the concentration (Gini) of events across trees serve as proxies.

Validity checks. (1) Stability of slopes between 2020–2022 and 2023–2025 and across the two drought summers. (2) Raw within-year heat–watering plots before modelling. (3) Discriminant check: if slopes are almost perfectly predicted by turnout, they measure general civic engagement rather than heat response, and the interpretation is narrowed accordingly.

Interpretive limits. The slope is a descriptive behavioral response, not a causal effect of weather on civic action, and not a measure of residents' own protection from heat. Area-level results describe places, not individual volunteers.

6.5 Part III — Allocation around volunteer effort (exploratory)

Population and label. Tree-seasons of catalogued street trees aged 5–19 years (all catalogued trees as sensitivity). The label is any logged watering in that season. Restricting to young street trees raises prevalence well above the 1.2% of the full catalogue and focuses on trees past municipal establishment care.

Features. Genus and drought sensitivity class, age, imperviousness and land use at the tree, local LST and canopy, planning-area socio-demographics, and neighbor-based features (e.g. watering within 100

m in the previous season). Neighbor features are computed **inside each fold from training data only**, to prevent leakage.

Models and validation. Regularized logistic regression (interpretable baseline) and gradient boosting. Validation uses planning-area-blocked cross-validation and a temporal hold-out (train 2020–2023, test 2024–2025). Discrimination is reported with AUROC and PR-AUC; probabilities are calibrated with isotonic regression and checked with reliability diagrams and Brier scores (Niculescu-Mizil and Caruana, 2005). A district hold-out tests portability.

Decision layer. A product score Need × (1 − $\hat{P}$) would collapse to Need when $\hat{P}$ is small for most trees. The thesis instead uses **expected unmet water**:

$$U(i,s) = \max\{0,\ R(i) - \hat{P}(i,s) \cdot \bar{L}(i)\}, \quad Priority(i,s) = N(i) \cdot U(i,s)$$

where R is the recommended seasonal dose by age class, $\bar{L}$ the expected logged litres given watering, and N a tree-level need term (genus drought sensitivity, sealing, age), optionally weighted by planning-area heat and social status from Part I. If QTrees water-need estimates become available, they replace R. Need-only and complement-based allocations are compared at planning-area level by Spearman rank correlation and top-50 overlap; if the two rankings are nearly identical, that is reported as a finding. Weight sensitivity is reported in full, and the score is framed as decision support, not an effect estimate.

Interpretive limit. The model predicts **logged** care. Trees without a log are described as having no recorded watering. Where district schedules are not obtained, trees under municipal care cannot be identified, which is why the population starts at age 5.

6.6 Integration and the normative choice

Part I supplies total cooling loss and residual thermal load to the bridge test (H4) and, optionally, to the need weighting in Part III. Part II supplies civic heat response as a separate evidence layer. Civic response is **not** folded into need: doing so would give less public service to areas because their residents act. Need-only and complement-based allocations are reported side by side, and the choice between them is treated as a policy question.

6.7 Robustness and multiplicity

Confirmatory testing is limited to H1, H2/H2-alt, H3 and H4. Secondary coefficients are grouped by family and controlled with the Benjamini–Hochberg procedure. All area results are repeated at three scales, and analyses are rerun excluding areas with weak registry coverage.

7. Expected Contributions

- **Civic heat response (primary).** A daily, city-wide behavioral indicator of collective response to heat, with an explicit test of whether its social gradient reflects vulnerability or participation inequality, and a benchmark against static indices.
- **European evidence on unequal cooling loss.** The first within-city European estimate of drought-driven cooling loss by planning area, separating canopy from low vegetation and extending US findings (Yan et al., 2025).
- **An allocation comparison around volunteer effort.** A leakage-resistant, calibrated model of logged care and a transparent comparison of need-only and complement-based allocations, designed to plug into sensor-based water-need estimates.
- **Reproducible pipeline.** Open code for integrating Landsat, Sentinel-2, registry and citizen-log data, reusable in other cities running Gieß den Kiez.

Long-term relevance. Higher-resolution thermal missions (TRISHNA, ~57 m, launch planned 2026; Copernicus LSTM, 50 m, planned ~2028) will allow Part I to be re-estimated with midday, more frequent data; the code is written to accept them. Part II's behavioral approach does not depend on satellite resolution, and transfers to other cities that run the same platform.

8. Work Plan and Timeline

Phase	Period	Tasks	Deliverable / decision
1 — Setup and audit	Month 1	Structured literature search (Gap 3); data audit and count reconciliation; data requests; pre-registration of H1–H4	Protocol, OSF pre-registration. **Decision:** offset definition; user-ID analysis on or proxy
2 — Panels	Months 1–2	Area-day behavioral panel; Landsat–Sentinel–vegetation-height panel; weather and calendar variables	Two analysis-ready panels
3 — Part II (core)	Months 2–4	Two-part model; fast fits then brms; H2 vs H2-alt; H3; stability checks	Civic heat-response layer; draft core paper
4 — Part I	Months 3–4	One-stage LST model; canopy/low-vegetation split; placebo; ECOSTRESS check	Cooling-loss map. **Decision (end Month 4):** Part III full or descriptive
5 — Part III	Month 5	Tree-season model; blocked and temporal validation; calibration; allocation comparison	Allocation comparison and ranked lists
6 — Synthesis	Month 6	H4 bridge test; MAUP appendix; writing; code release	Submitted thesis

9. Risks and Mitigation

Risk	Likelihood	Mitigation
Brms models too slow at planning-area level	Medium	Bezirksregion fallback; glmmTMB/INLA for exploration; cluster time booked in Month 1
No user IDs or district schedules	High	Proxy analyses (distinct trees, Gini; age ≥ 5 population) specified in advance
Weak heat signal in watering after year and weekday controls	Low–medium	Report as a finding; shift emphasis to participation (part a) and persistence
Part I dry/wet contrast confounded by date conditions	Medium	Date fixed effects; placebo on sealed cells; continuous anomaly
Time overrun	Medium	Part III can be reduced to a descriptive comparison without affecting the core study

10. Expected Outputs and Publication Strategy

- **Thesis.** About 70–80 pages: background, three results chapters (core first), synthesis with the bridge test, limitations, policy notes, and appendices (scale analysis, MAUP, diagnostics).
- **Atlas and data layers.** Planning-area maps of cooling loss and civic heat response with uncertainty, and allocation rankings with an indicator dictionary.
- **Code.** Version-controlled pipeline and documentation.
- **Primary paper (Part II).** Civic heat response and participation inequality, targeted at Global Environmental Change, Landscape and Urban Planning or npj Urban Sustainability.
- **Optional second paper (Part I).** Unequal drought-driven cooling loss in a European city (Urban Climate), if the canopy/low-vegetation split yields a clear result.

11. Ethical Considerations and Limitations

11.1 Data ethics

Public logs record tree IDs, dates, locations and volumes, not volunteer identities. If pseudonymous user IDs are provided, they are used only in aggregate, never linked to other data, and stored under a

data-use agreement. Outputs avoid maps or tables that could re-identify individuals, for example areas with a single active volunteer. Socio-economic data are used only in published aggregate form. Ethics requirements will be confirmed with the university before analysis.

Part III raises a fairness issue: allocating public irrigation away from areas where residents volunteer shifts a maintenance burden onto them. Both allocations are therefore reported, and neither is recommended as technically correct.

11.2 Methodological limitations

- **Surface, not felt heat.** LST at ~10:30 is not near-ground air temperature or night-time heat, which matter most for health.
- **Logged care is not care.** Unlogged and municipal watering are unobserved.
- **Descriptive response.** Civic heat-response slopes are not causal and not a direct measure of adaptive capacity.
- **Clear-sky selection and coarse drought.** Part I covers fair-weather days and uses a city-level drought series.
- **Single-date vegetation height and dated built-form layers.** Canopy change over 2017–2024 is captured only through NDVI.
- **Registry coverage.** Green-space trees are incompletely catalogued; the register has no vitality field.
- **Scale.** Results may change with aggregation; they are reported at three scales.

Appendix A. Why a Satellite Test of Watering on Temperature Is Not Attempted

Dose. Guidance for a young Berlin street tree is about 500 L per growing season. The observed median is 65 L per tree-season; 42.0% of logged tree-seasons reach 100 L, 15.9% reach 250 L and 6.3% reach 500 L.

Spatial concentration. On a 30 m grid, 7,174 cells contain watered trees, with a median of one per cell (90th percentile 3, maximum 11). On a 100 m grid, 4,205 cells, median two (90th percentile 6, maximum 29).

Energy budget (corrected). With latent heat 2.45 MJ kg^{-1} and a combined surface coupling of about 20 W m^{-2} K^{-1}, the implied surface-temperature change if 30 L evaporates within the stated window is:

Case	If evaporated over 6 h	If evaporated over 24 h
30 L over one 100 m cell	0.34 W m^{-2} → 0.017 K	0.085 W m^{-2} → 0.004 K
30 L over one 30 m cell	3.8 W m^{-2} → 0.19 K	0.95 W m^{-2} → 0.047 K
Densest 100 m cell (29 trees, one day)	→ 0.49 K	→ 0.12 K

The earlier version reported the 24-hour values under a six-hour label. Both are upper bounds: most poured water infiltrates and is transpired over days, so the instantaneous latent-heat flux at the ~10:30 overpass is smaller still.

Detectability. Landsat C2 L2 LST has 1–2 K RMSE against ground truth, but much of that is bias that cancels in within-pixel comparisons, so the relevant noise is the pixel-level precision of a change. Even assuming an optimistic 0.3 K, a 0.05 K effect would require roughly 280 independent paired treated–control observations at matched times. Individual cells offer only 10–20 clear-sky summer overpasses per year, watering is rarely timed just before an overpass, and watered trees are not randomly chosen.

Conclusion (scoped). At observed watering densities, the effect is not recoverable from Landsat-class data without a designed experiment that concentrates watering and controls its timing. This conclusion may change with near-daily 50–60 m thermal data, and is presented as a thesis appendix rather than a standalone publication.

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Approval and Signatures

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